SoccerGMR Standard Test (1036/1036) — primary

Standard-trained checkpoints | Standard Test (1036/1036) | 544 positive / 492 negative

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	AUROC	mAP	mR@1	mR@5	mR+@5	N-acc τ=0.4	Rej-F1 τ=0.4	G-mIoU@1 τ=0.4	G-mIoU@3 τ=0.4	N-acc τ=0.5	Rej-F1 τ=0.5	G-mIoU@1 τ=0.5	G-mIoU@3 τ=0.5	N-acc τ=0.6	Rej-F1 τ=0.6	G-mIoU@1 τ=0.6	G-mIoU@3 τ=0.6
Moment-DETR	**GMR (baseline)**	**68.80**	**5.91**	**3.68**	**7.98**	**1.16**	**0.00**	**0.00**	**0.0577**	**0.0220**	**100.00**	**75.00**	**0.5239**	**0.4912**	**100.00**	**75.00**	**0.5239**	**0.4912**
	GMR + GEQ Frozen	68.18	5.91	3.68	7.98	1.16	100.00	75.00	0.5239	0.4912	100.00	75.00	0.5239	0.4912	100.00	75.00	0.5239	0.4912
		(-0.62)	(0.00)	(0.00)	(0.00)	(0.00)	(+100.00)	(+75.00)	(+0.4662)	(+0.4692)	(0.00)	(0.00)	(0.0000)	(0.0000)	(0.00)	(0.00)	(0.0000)	(0.0000)
	GMR + GEQ Full	68.96	6.09	3.77	8.19	0.66	99.59	74.81	0.5243	0.4901	100.00	75.00	0.5263	0.4920	100.00	75.00	0.5263	0.4920
		(+0.16)	(+0.18)	(+0.09)	(+0.21)	(-0.50)	(+99.59)	(+74.81)	(+0.4666)	(+0.4681)	(0.00)	(0.00)	(+0.0024)	(+0.0008)	(0.00)	(0.00)	(+0.0024)	(+0.0008)
	GMR + Multi	70.14	7.27	4.71	10.04	0.28	83.54	69.60	0.4421	0.4136	100.00	75.00	0.5188	0.4895	100.00	75.00	0.5188	0.4895
		(+1.34)	(+1.36)	(+1.03)	(+2.06)	(-0.88)	(+83.54)	(+69.60)	(+0.3844)	(+0.3916)	(0.00)	(0.00)	(-0.0051)	(-0.0017)	(0.00)	(0.00)	(-0.0051)	(-0.0017)
	GMR + Multi + GEQ Frozen	69.25	7.27	4.71	10.04	0.28	90.65	72.11	0.4749	0.4456	100.00	75.00	0.5188	0.4895	100.00	75.00	0.5188	0.4895
		(+0.45)	(+1.36)	(+1.03)	(+2.06)	(-0.88)	(+90.65)	(+72.11)	(+0.4172)	(+0.4236)	(0.00)	(0.00)	(-0.0051)	(-0.0017)	(0.00)	(0.00)	(-0.0051)	(-0.0017)
	GMR + Multi + GEQ Full	69.83	7.72	4.06	12.48	0.83	81.91	68.89	0.4395	0.4070	100.00	75.00	0.5237	0.4912	100.00	75.00	0.5237	0.4912
		(+1.03)	(+1.81)	(+0.38)	(+4.50)	(-0.33)	(+81.91)	(+68.89)	(+0.3818)	(+0.3850)	(0.00)	(0.00)	(-0.0002)	(0.0000)	(0.00)	(0.00)	(-0.0002)	(0.0000)
EaTR	**GMR (baseline)**	**76.56**	**16.20**	**10.64**	**21.38**	**9.59**	**48.58**	**57.11**	**0.3332**	**0.2834**	**66.67**	**65.60**	**0.4085**	**0.3600**	**87.20**	**73.46**	**0.4956**	**0.4486**
	GMR + GEQ Frozen	75.17	16.20	10.64	21.38	9.59	63.82	64.02	0.3911	0.3452	87.60	72.93	0.4933	0.4489	96.75	74.90	0.5315	0.4858
		(-1.39)	(0.00)	(0.00)	(0.00)	(0.00)	(+15.24)	(+6.91)	(+0.0579)	(+0.0618)	(+20.93)	(+7.33)	(+0.0848)	(+0.0889)	(+9.55)	(+1.44)	(+0.0359)	(+0.0372)
	GMR + GEQ Full	75.26	17.15	11.47	22.84	10.79	64.23	64.29	0.3988	0.3482	90.65	74.77	0.5111	0.4635	98.17	75.23	0.5402	0.4921
		(-1.30)	(+0.95)	(+0.83)	(+1.46)	(+1.20)	(+15.65)	(+7.18)	(+0.0656)	(+0.0648)	(+23.98)	(+9.17)	(+0.1026)	(+0.1035)	(+10.98)	(+1.77)	(+0.0446)	(+0.0435)
	GMR + Multi	77.11	17.52	11.64	23.70	11.77	44.92	55.18	0.3305	0.2702	61.59	63.79	0.3988	0.3424	75.41	69.15	0.4575	0.4012
		(+0.55)	(+1.32)	(+1.00)	(+2.32)	(+2.18)	(-3.66)	(-1.93)	(-0.0027)	(-0.0132)	(-5.08)	(-1.81)	(-0.0097)	(-0.0176)	(-11.79)	(-4.31)	(-0.0381)	(-0.0474)
	GMR + Multi + GEQ Frozen	76.69	17.52	11.64	23.70	11.77	69.51	68.13	0.4277	0.3738	81.50	71.04	0.4738	0.4243	89.63	73.01	0.5058	0.4571
		(+0.13)	(+1.32)	(+1.00)	(+2.32)	(+2.18)	(+20.93)	(+11.02)	(+0.0945)	(+0.0904)	(+14.84)	(+5.44)	(+0.0653)	(+0.0643)	(+2.44)	(-0.45)	(+0.0102)	(+0.0085)
	GMR + Multi + GEQ Full	77.27	19.88	14.05	25.31	12.24	68.90	67.60	0.4318	0.3741	81.30	71.05	0.4796	0.4260	90.65	73.48	0.5165	0.4632
		(+0.71)	(+3.68)	(+3.41)	(+3.93)	(+2.65)	(+20.33)	(+10.49)	(+0.0986)	(+0.0907)	(+14.63)	(+5.45)	(+0.0711)	(+0.0660)	(+3.46)	(+0.02)	(+0.0209)	(+0.0146)
QD-DETR	**GMR (baseline)**	**77.66**	**15.08**	**10.04**	**20.11**	**9.62**	**53.46**	**59.84**	**0.3533**	**0.3035**	**90.04**	**73.77**	**0.5058**	**0.4611**	**97.56**	**75.29**	**0.5355**	**0.4901**
	GMR + GEQ Frozen	77.42	15.08	10.04	20.11	9.62	77.24	70.11	0.4559	0.4070	92.48	74.47	0.5161	0.4711	96.95	74.88	0.5322	0.4868
		(-0.24)	(0.00)	(0.00)	(0.00)	(0.00)	(+23.78)	(+10.27)	(+0.1026)	(+0.1035)	(+2.44)	(+0.70)	(+0.0103)	(+0.0100)	(-0.61)	(-0.41)	(-0.0033)	(-0.0033)
	GMR + GEQ Full	77.66	16.54	11.48	22.09	10.14	80.49	71.35	0.4727	0.4232	93.90	74.52	0.5241	0.4769	98.17	75.23	0.5419	0.4942
		(0.00)	(+1.46)	(+1.44)	(+1.98)	(+0.52)	(+27.03)	(+11.51)	(+0.1194)	(+0.1197)	(+3.86)	(+0.75)	(+0.0183)	(+0.0158)	(+0.61)	(-0.06)	(+0.0064)	(+0.0041)
	GMR + Multi	75.56	14.67	9.95	19.50	5.60	53.86	58.37	0.3485	0.2985	91.67	74.92	0.5157	0.4690	96.54	75.28	0.5308	0.4846
		(-2.10)	(-0.41)	(-0.09)	(-0.61)	(-4.02)	(+0.41)	(-1.47)	(-0.0048)	(-0.0050)	(+1.63)	(+1.15)	(+0.0099)	(+0.0079)	(-1.02)	(-0.01)	(-0.0047)	(-0.0055)
	GMR + Multi + GEQ Frozen	74.92	14.67	9.95	19.50	5.60	74.59	68.41	0.4380	0.3917	94.51	74.82	0.5224	0.4772	97.97	74.67	0.5320	0.4886
		(-2.74)	(-0.41)	(-0.09)	(-0.61)	(-4.02)	(+21.14)	(+8.57)	(+0.0847)	(+0.0882)	(+4.47)	(+1.05)	(+0.0166)	(+0.0161)	(+0.41)	(-0.62)	(-0.0035)	(-0.0015)
	GMR + Multi + GEQ Full	74.79	13.71	9.21	18.42	6.44	94.31	74.72	0.5194	0.4756	98.98	75.15	0.5368	0.4932	99.59	74.87	0.5387	0.4951
		(-2.87)	(-1.37)	(-0.83)	(-1.69)	(-3.18)	(+40.85)	(+14.88)	(+0.1661)	(+0.1721)	(+8.94)	(+1.38)	(+0.0310)	(+0.0321)	(+2.03)	(-0.42)	(+0.0032)	(+0.0050)
CG-DETR	**GMR (baseline)**	**75.97**	**16.09**	**10.76**	**22.12**	**9.48**	**96.75**	**75.68**	**0.5382**	**0.4890**	**99.19**	**75.84**	**0.5465**	**0.4975**	**99.59**	**75.91**	**0.5484**	**0.4994**
	GMR + GEQ Frozen	76.39	16.09	10.76	22.12	9.48	75.00	69.04	0.4457	0.3969	92.48	74.23	0.5189	0.4698	99.19	75.95	0.5463	0.4976
		(+0.42)	(0.00)	(0.00)	(0.00)	(0.00)	(-21.75)	(-6.64)	(-0.0925)	(-0.0921)	(-6.71)	(-1.61)	(-0.0276)	(-0.0277)	(-0.41)	(+0.04)	(-0.0021)	(-0.0018)
	GMR + GEQ Full	74.83	16.58	11.12	22.60	10.97	75.00	68.59	0.4439	0.3933	97.36	75.55	0.5381	0.4889	99.80	75.42	0.5446	0.4978
		(-1.14)	(+0.49)	(+0.36)	(+0.48)	(+1.49)	(-21.75)	(-7.09)	(-0.0943)	(-0.0957)	(-1.83)	(-0.29)	(-0.0084)	(-0.0086)	(+0.20)	(-0.49)	(-0.0038)	(-0.0016)
	GMR + Multi	77.40	16.59	11.21	21.77	8.14	46.95	56.20	0.3369	0.2783	75.41	69.61	0.4542	0.4005	90.45	74.23	0.5167	0.4641
		(+1.43)	(+0.50)	(+0.45)	(-0.35)	(-1.34)	(-49.80)	(-19.48)	(-0.2013)	(-0.2107)	(-23.78)	(-6.23)	(-0.0923)	(-0.0970)	(-9.15)	(-1.68)	(-0.0317)	(-0.0353)
	GMR + Multi + GEQ Frozen	75.95	16.59	11.21	21.77	8.14	70.33	67.32	0.4310	0.3758	85.16	72.68	0.4938	0.4417	92.68	73.91	0.5198	0.4701
		(-0.02)	(+0.50)	(+0.45)	(-0.35)	(-1.34)	(-26.42)	(-8.36)	(-0.1072)	(-0.1132)	(-14.02)	(-3.16)	(-0.0527)	(-0.0558)	(-6.91)	(-2.00)	(-0.0286)	(-0.0293)
	GMR + Multi + GEQ Full	75.77	18.59	12.31	24.98	12.13	59.15	62.72	0.3884	0.3304	81.30	71.43	0.4784	0.4246	93.90	74.04	0.5244	0.4750
		(-0.20)	(+2.50)	(+1.55)	(+2.86)	(+2.65)	(-37.60)	(-12.96)	(-0.1498)	(-0.1586)	(-17.89)	(-4.41)	(-0.0681)	(-0.0729)	(-5.69)	(-1.87)	(-0.0240)	(-0.0244)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.